

Lab Azure Container Apps – Deploying the Database Container To Begin with the Lab

Summary

In this lab, we deploy a **MySQL database container** as a separate **Azure Container App** within the same **Container Apps Environment** as the web application. The database container is configured with environment variables, exposed only internally using **TCP port 3306**, and accessed securely by the web application.

Understanding

Azure Container Apps allow multiple containers to run inside the same **Container Apps Environment**. In this lab, one Container App hosts the **ASP.NET web application**, while another hosts the **MySQL database**. Since both apps are in the same environment, the web application can communicate with the database without exposing it to the public internet. Container Apps also support **automatic scaling**. By default, if there is no incoming traffic, a Container App can scale down to **zero replicas**, reducing costs. If continuous availability is required, the minimum replica count can be set to **1**.

The MySQL container requires the **MYSQL_ROOT_PASSWORD** environment variable during deployment. Unlike the web application, the database uses **TCP traffic on port 3306** instead of HTTP.

Example:

- **Container App 1:** ASP.NET Web Application (HTTP/HTTPS)
- **Container App 2:** MySQL Database (TCP 3306)
- Both run in the same **Container Apps Environment** and communicate internally.

Lab Steps

Step 1: Open the Existing Container App

- Open the Azure Portal.
- Navigate to the deployed **Container App**.
- Open **Revisions and Replicas**.
- Verify the application revision.

The screenshot shows the Azure Portal interface for a Container App. The breadcrumb navigation at the top reads: Home > MicrosoftApp-ContainerApp-Portal-1c391ab8-aae0 | Overview > cloudxeuss-webapp. The main header shows the app name 'cloudxeuss-webapp' with a 'Container App' icon, followed by 'Revisions and replicas' and a star icon. A left sidebar contains navigation links: Overview, Activity log, Access control (IAM), Tags, Diagnose and solve problems, and Resource visualizer. Under 'Application', there are links for Revisions and replicas (highlighted with a red box), Containers, Scale, and Volumes. The main content area has a search bar and a message about the new revision mode. Below this is a table with tabs for 'Active revisions', 'Inactive revisions', and 'Replicas'. The table has columns: Name, Date created, Running status, View Logs, Label, Traffic, and Replicas. One revision is listed: 'cloudxeuss-webapp--r1yh3b' with a date of '7/24/2026, 10:44:14 AM', a 'Running' status with a green checkmark, and 1 replica. The 'Replicas' column shows '1 (Show replicas)'.

Name	Date created	Running status	View Logs	Label	Traffic	Replicas
cloudxeuss-webapp--r1yh3b	7/24/2026, 10:44:14 AM	Running View detail Show Logs			100 %	1 (Show replicas)

Step 2: Check Replica Status

- Observe the running replicas.
- If there is no traffic, the Container App may show **0 running instances**.

- This is normal because Container Apps automatically scale down to save cost.

The screenshot shows the 'Revision details' page for a container app named 'cloudxeuss-webapp'. The 'Running status' is highlighted with a red box and shows 'Scaled to 0'. Other details include: Revision name: cloudxeuss-webapp--rf1yh3b, Revision URL: cloudxeuss-webapp--rf1yh3b.yellowhill-e631480b.eastus.azurecontainerapps.io, Status: Active, Time created: 7/24/2026, 10:44:14 AM, Traffic: 100%, Active/total replicas: No revision replicas (Show details), and Min-max replicas: 0 - 10 (Show details).

Step 3: Configure Minimum Replicas

- Click **Show Details** under Min/Max Replicas.

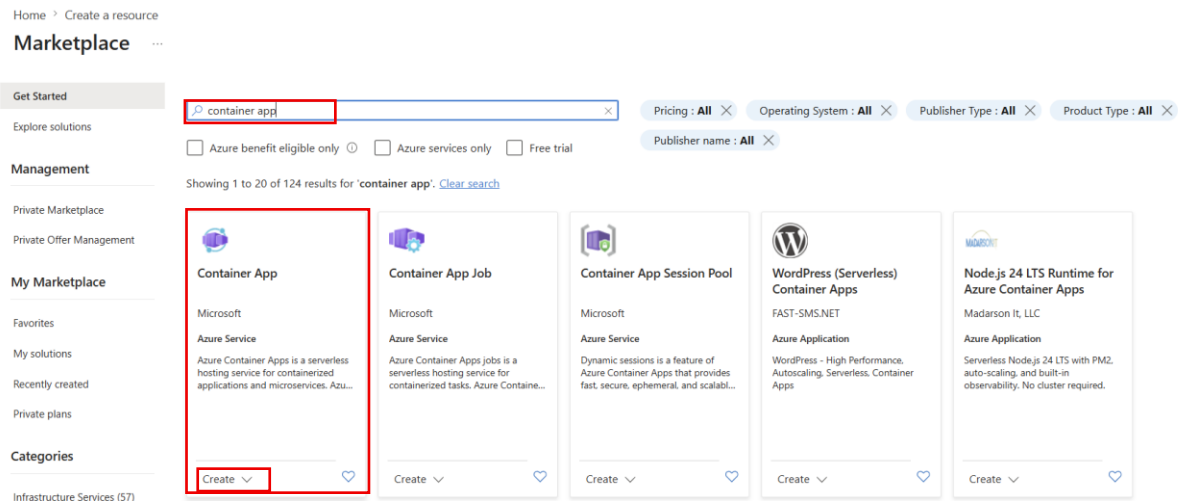
This screenshot is similar to the previous one but highlights the 'Min-max replicas' field with a red box, which shows '0 - 10 (Show details)'. The 'Running status' is also visible as 'Scaled to 0'.

- Change **Minimum Replicas** from **0** to **1**.
- Save the configuration.
- Azure creates a **new revision**.
- Wait until the revision becomes active.

The screenshot shows the 'Scale' configuration page for the container app. The 'Based on revision' dropdown is set to 'cloudxeuss-webapp--rf1yh3b'. Under 'Scale rule settings', the 'Min replicas' is set to 1, 'Max replicas' is 10, 'Cooldown period' is 300, and 'Polling interval' is 30. The 'Current number of replicas' is 0. At the bottom, the 'Save as a new revision' button is highlighted with a red box.

Step 4: Create a New Container App for MySQL

- Go to **Azure Portal**.
- Search for **Container Apps**.
- Click **Create**.



- Select the existing **Resource Group**.
- Enter the **Container App Name**
- Select **East US Region**.
- Use the same **Container Apps Environment** created earlier.

A screenshot of the 'Create Container app' form in the Azure Portal. The form is titled 'Create Container app'. It has several fields and sections: 'Resource group *' is set to 'Webapp'; 'Container app name *' is set to 'container-app'; 'Optimize for Azure Functions' is unchecked; 'Built-in support and autoscaling for Azure Functions' is checked; 'Deployment source *' is set to 'Container image'; 'Container Apps environment' is set to 'managedEnvironment-DefaultResource-bcce (DefaultResourceGroup-EUS)'; 'Region *' is set to 'East US'. The 'Home' link at the top left is highlighted with a red box. The 'Resource group *' and 'Container app name *' fields are highlighted with red boxes. The 'Container Apps environment' field is highlighted with a red box.

Step 5: Select the MySQL Container Image

- Click **Next: Container**.
- Choose the **Azure Container Registry (ACR)**.
- Select the **MySQL Docker Image**.
- Select the required **Image Tag**.

Home

Create Container app ...

Basics **Container** Ingress Tags Review + create

Select a quickstart image for your container, or deselect quickstart image to use an existing container.

Use quickstart image

☐

Container details

Name *

container-app

Image source

☒ Azure Container Registry

☐ Docker Hub or other registries

Subscription *

Visual Studio Enterprise Subscription

Registry *

regdevs12.azurecr.io

Image *

cloudxues-mysql

Image tag *

8.4

Step 6: Configure Environment Variables

- Add the required environment variable.

Variable

Value

MYSQL_ROOT_PASSWORD Mysql@123

- Save the environment variable.

Container resource allocation

Choose the workload profile for this app. You can adjust the CPU and memory allocation for this app up to the workload profile limit. [Learn more](#)

Workload profile *

Consumption - Up to 4 vCPUs, 8 Gib memory

GPU

☐

CPU and memory *

0.5 CPU cores, 1 Gi memory

Environment variables

Name	Value	Delete
MYSQL_ROOT_PASSWORD	Mysql@123	
Enter name	Enter value	

[Review + create](#)

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[Next : Ingress >](#)

Step 7: Configure Ingress

- Enable **Ingress**.
- Choose **Limited to Container Apps Environment (Internal)**.
- Select **TCP** as the transport protocol.
- Set **Target Port** to **3306**.
- Do **not** use HTTP because MySQL communicates using TCP.

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Create Container app ...

Basics Container Ingress Tags Review + create

Application ingress settings

Enable ingress for applications that need an HTTP or TCP endpoint.

Ingress  ☒

Ingress traffic

- ☒ Limited to Container Apps Environment
Select this option if you want to restrict traffic to this container app from within the Container App Environment
- ☐ Accepting traffic from anywhere
Select this option if you want to allow traffic to this container app from anywhere

Ingress type  ☐ HTTP ☒ TCP

Target port 

Exposed port 

 Additional TCP ports

Review + create

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Next : Tags >

Step 8: Deploy the Database Container

- Click **Review + Create**.
- Click **Create**.

Home

Create Container app ...

✓ Passed

Basics Container Ingress Tags **Review + create**

Project details

Subscription Visual Studio Enterprise Subscription
Resource group Webapp
Name container-app

Container Apps Environment

Region eastus
Container Apps Environment managedEnvironment-DefaultResource-bcce

Container

Name container-app
Image source Azure Container Registry
Registry regdeves12.azurecr.io
Image cloudevents-mysql
Image tag 8.4
Command
Arguments
Workload profile type Consumption

Create

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- Wait until deployment completes successfully.

Home

Microsoft.App-ContainerApp-Portal-7b67a02b-a2ad | Overview ...

Deployment

Search x « Delete Cancel Redeploy Download Refresh

Overview

Inputs

Outputs

Template

✓ **Your deployment is complete**

Deployment name : Microsoft.App-ContainerApp-Portal-7b67a02b-a2ad
Subscription : Visual Studio Enterprise Subscription
Resource group : Webapp

Start time : 7/24/2026, 2:54:26 PM
Correlation ID : 3019d3fd-1121-4099-ba77-766348e344e4

> Deployment details

Next steps

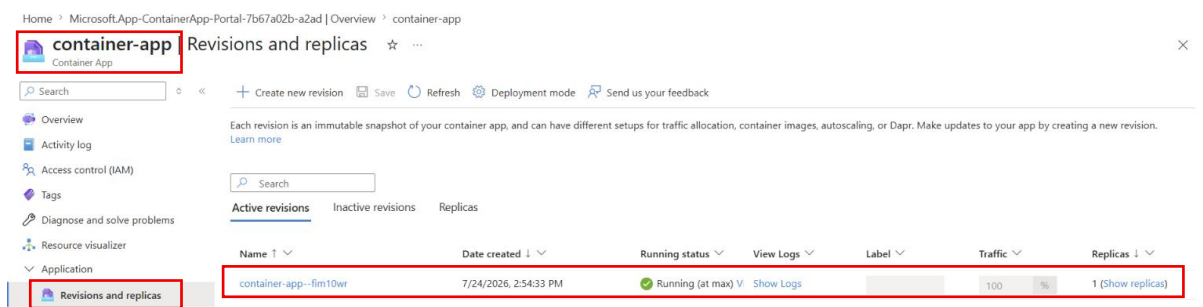
[Go to resource](#)

Give feedback

[Tell us about your experience with deployment](#)

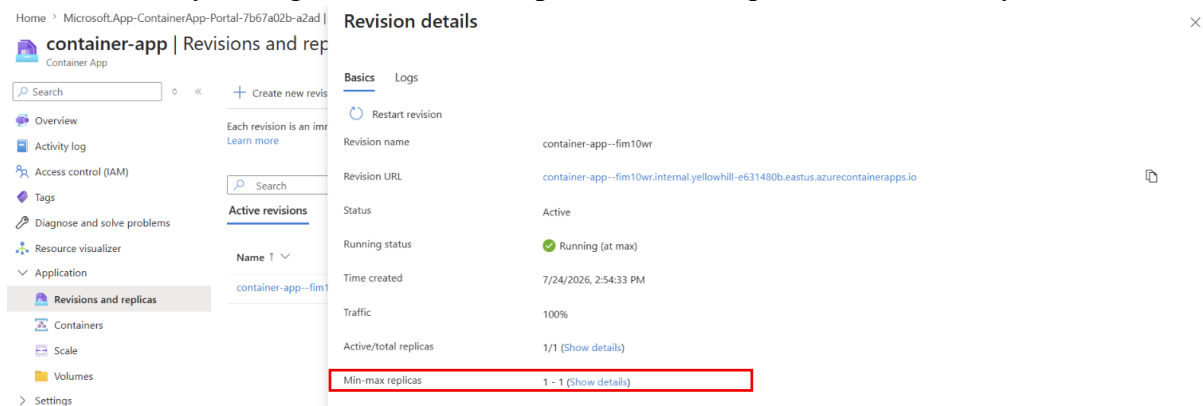
Step 9: Verify Deployment

- Open the newly created **Container App**.
- Navigate to **Revisions and Replicas**.
- Verify that the revision is active.
- Refresh the page if required.



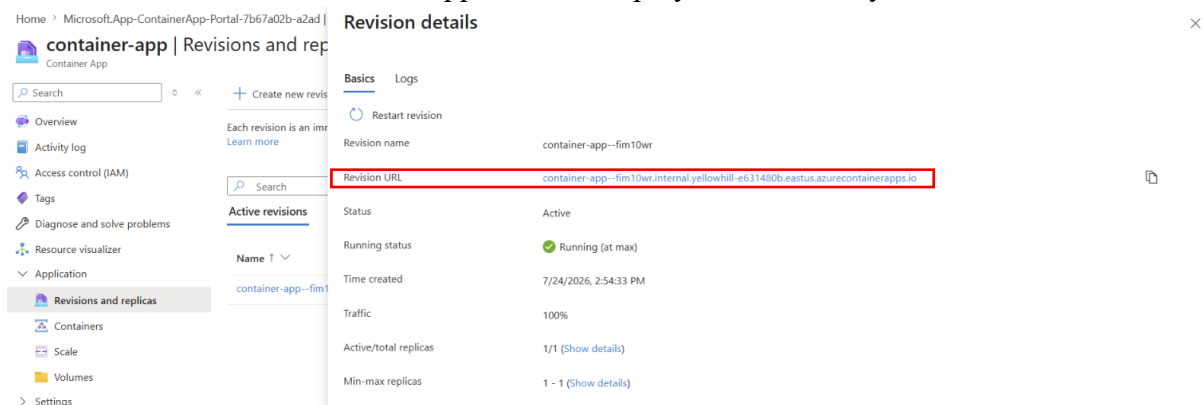
Step 10: Verify Replica Status

- Ensure the MySQL Container App is running.
- If necessary, configure **Minimum Replicas = 1** to keep one instance always active.



Step 11: Copy the Revision URL

- Open the active revision.
- Copy the generated **Revision URL**.
- This confirms the Container App has been deployed successfully.



Step 12: Test the Application

- Open the **Orders** page of the web application.
- Verify that the application retrieves data successfully.
- Confirm the web application can communicate with the MySQL database.

Course Orders

OrderId	Customer	Email	Course	Amount	OrderDateUtc
5	Customer 005	customer005@cloudxeus.com	AI-9004: Azure AI Fundamentals	19.00	2026-07-22 10:34:41Z
4	Customer 004	customer004@cloudxeus.com	AZ-3056: Azure Architect	69.00	2026-07-22 10:34:41Z
3	Customer 003	customer003@cloudxeus.com	DP-6004: Fabric Analytics Engineer	59.00	2026-07-22 10:34:41Z
2	Customer 002	customer002@cloudxeus.com	AZ-1046: Azure Administrator	39.00	2026-07-22 10:34:41Z
1	Customer 001	customer001@cloudxeus.com	AZ-2045: Azure Developer Associate	49.00	2026-07-22 10:34:41Z